

Strategy 3

Parameter Ranges

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What varies, and what stays fixed

2

Fixed held constant

θ_r set to 0

Geometry

height 5.5 m, area 308 m², 25 days

Solver grid

Dz = 2.5 cm, Dt = 1/24 day

Varying · soil

$\Delta\theta$ drainable capacity ($\theta_s - \theta_r$)

n retention curve shape

α inverse air-entry

K_s saturated conductivity

Under VGM, these set:

- how much water the bed stores and drains
- the shape of the retention curve
- how fast water flows

Varying · operating conditions

$S_{e,i}$ initial condition

R/K_s irrigation relative to conductivity

Irrigation is **sampled as Q** but its **limits are set in R/K_s** , the ratio that governs flooding.

Soil parameters

3

VAN GENUCHTEN – MUALEM (VGM)

$$S_e = [1 + |\alpha h|^n]^{-m}, \quad m = 1 - 2/n$$
$$K(S_e) = K_s S_e^\delta, \quad \delta = 3 + 2/(m n)$$

EVERY BOUND IS ONE OF THREE KINDS

NUMERICAL set by the solver grid, not by physics

PHYSICAL a real soil or material limit

OPERATIONAL how the process actually runs, not a breaking point

$\Delta\theta$ drainable capacity, equals θ_s here ($\theta_r = 0$)

0.15 to 0.50

LOWER a stability limit on the solver grid (small capacity needs finer time steps)

UPPER a packed ore bed is not mostly open void

n shape of the VGM retention curve

2.2 to 5.0

LOWER structural floor: $n > 2$ keeps the conductivity term well posed

UPPER no physical upper limit; the ceiling is set by what the solver grid can resolve

α air-entry

0.005 to 0.05

LOWER the heavy-clay end of the literature range

UPPER no physical upper limit; set by grid resolution, since a higher α sharpens the front

K_s conductivity

170 to ~1360

LOWER anchored at 170 from the literature (Cariaga, Eppink); lower values are safe

UPPER explored to a rate limit near 1360 (about 8x); never binds in practice

Scenario inputs

4

$S_{e,i}$ initial wetness **0.40 to 0.95**

LOWER set just below the reference start (0.42)

UPPER clean to 0.95 even at strong forcing

Enters in the initial condition only. Per run: $\theta_i = S_{e,i} \cdot \Delta\theta$.

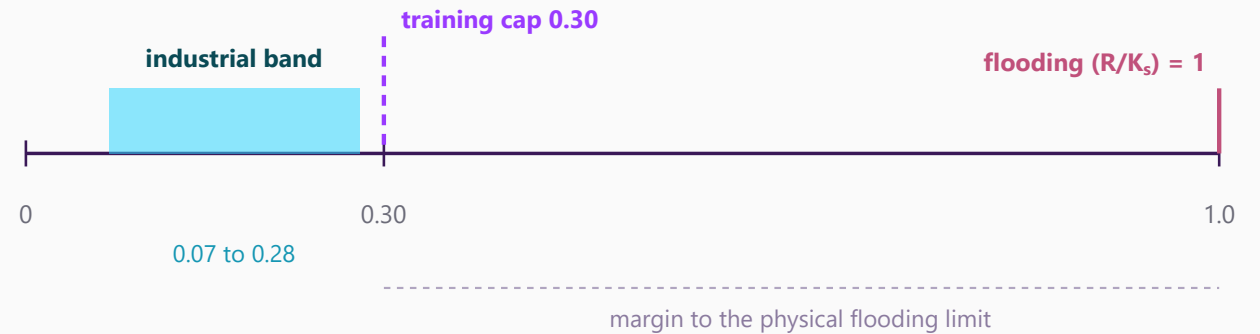
R/K_s irrigation forcing **~0.01 to 0.30**

UPPER 0.30 sits at the top of the industrial irrigation range reported in the literature

CONTEXT the physical flooding limit sits far above at $R/K_s = 1$

Per soil: $Q_{ub} = 0.30 \cdot K_s \cdot A / 100$.

Where the irrigation cap sits



Five converged sources put industrial irrigation at about 5 to 20 L/m²/h, which is R/K_s 0.07 to 0.28 at $K_s = 170$.

Each parameter's range is clean on its own

- **Alone, all clean**

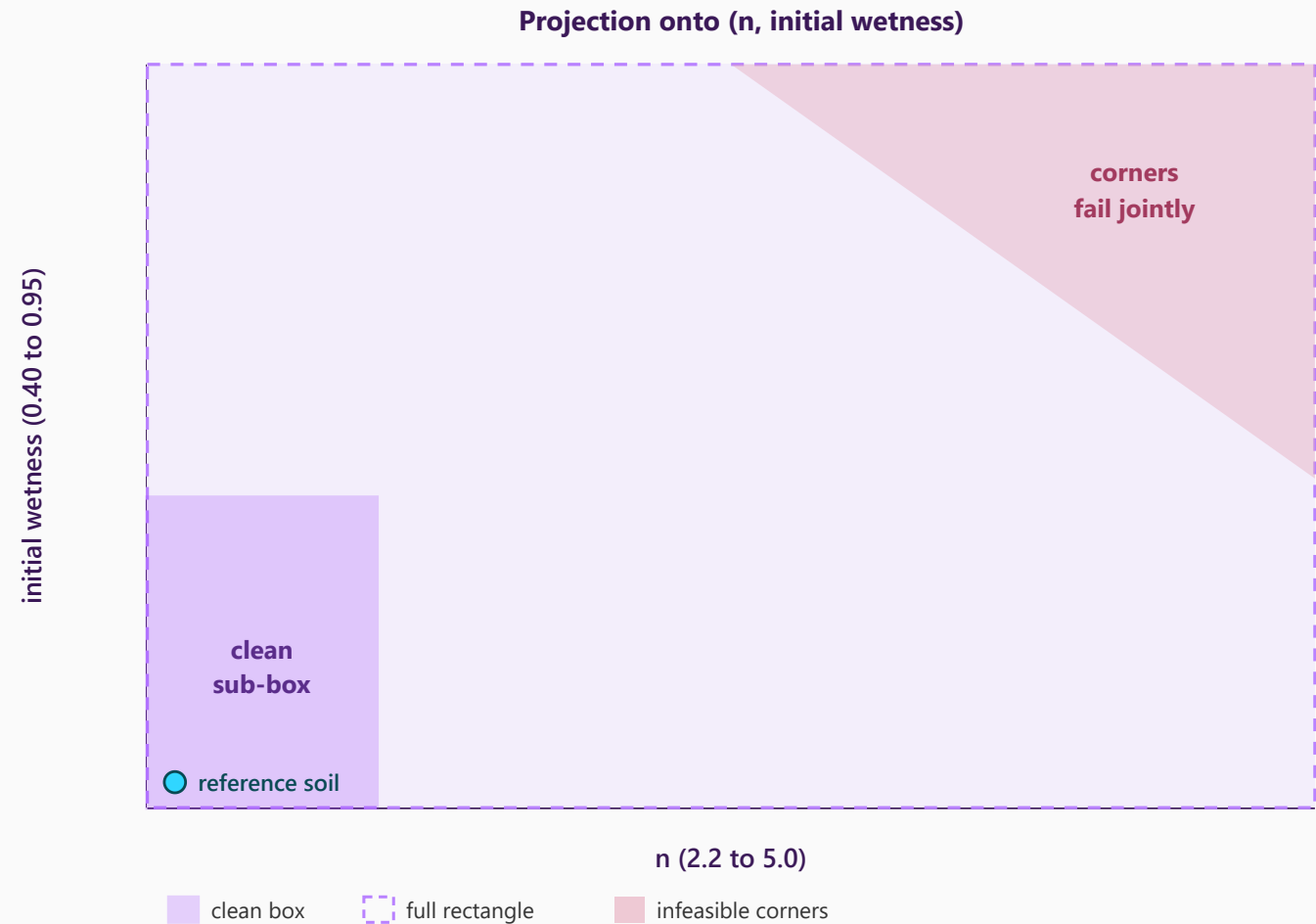
Pushed one parameter at a time across its full range, with the rest at reference, there are zero failures.

- **Together, some combinations fail**

The failures skew to the corner with high n and high initial wetness. A range that is safe on its own can still fail once combined with others.

- **So we cannot just take the box**

The usable training set is the clean region inside the rectangle.



The failures are all at the top surface

- **No single formula**

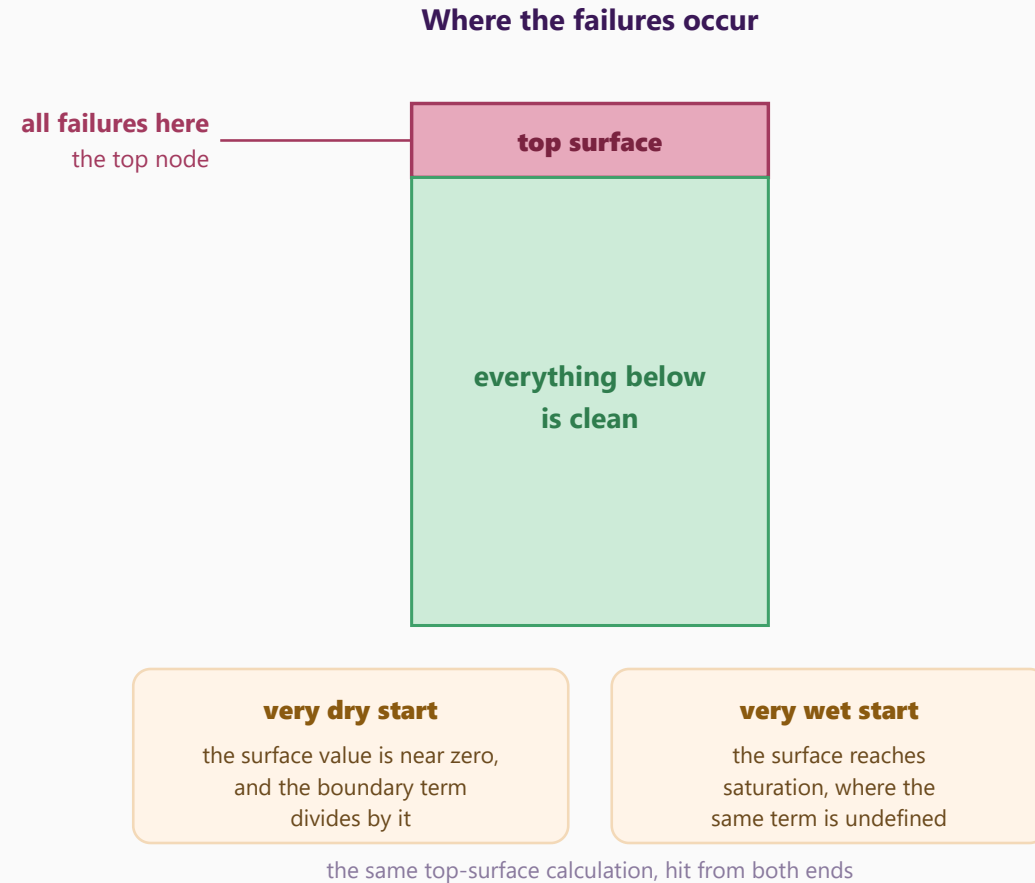
We hoped one number could flag the bad soils in advance. It cannot, so we measured where and why they fail.

- **Measurement showed**

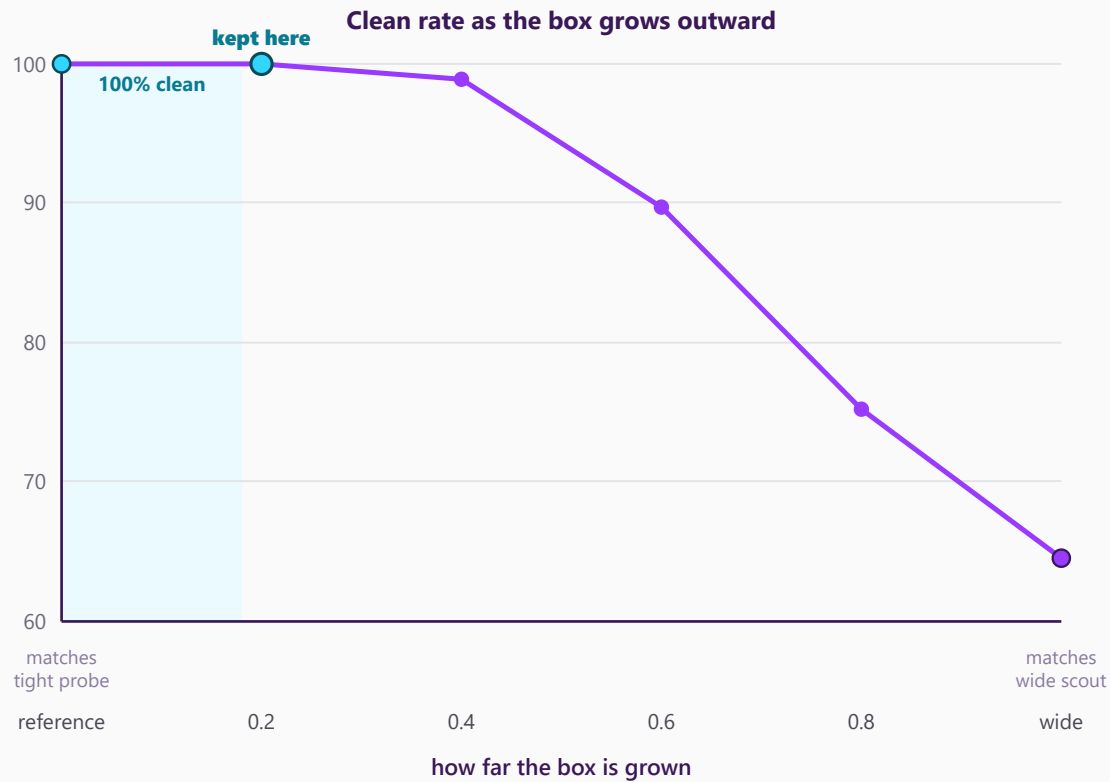
Every failure appears at the very top of the column, not deeper in the soil.

- **The box is clean below**

Away from the top surface there is no failure anywhere in the box. It is one boundary calculation, hit from two ends.



The largest feasible region



THE SEALED TRAINING BOX

K_s	130–232	n	2.20–3.00
α	0.025–0.042	$\Delta\theta$	0.270–0.388
$S_{e,i}$	0.40–0.63	R/K_s	≤ 0.30
per run: $\theta_i = S_{e,i} \cdot \Delta\theta$, $Q_{ub} = 0.30 \cdot K_s \cdot A / 100$			

Thank you

Questions welcome

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